

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Restaurant Management System

Version: 1.0

Project: Restaurant Management System

Document Type: Software Requirements Specification

1. Introduction

1.1 Purpose

The purpose of the Restaurant Management System is to computerize and simplify the daily operations of a restaurant. The system helps manage customers, food items, orders, billing, employees, tables, inventory, and reports.

This SRS document describes the functional and non-functional requirements of the Restaurant Management System.

1.2 Scope

The Restaurant Management System is designed to manage restaurant activities efficiently.

The system provides the following major functions:

- Customer management
- Table management
- Food/menu management
- Food ordering

- Order processing
- Billing and payment
- Inventory management
- Employee management
- Sales and report generation
- User login and security

The system reduces manual work and maintains restaurant information in a centralized database.

1.3 Definitions and Abbreviations

Term| Definition

RMS| Restaurant Management System

SRS| Software Requirements Specification

Admin| Person who manages the complete system

Customer| Person who orders food

Order| Request made by a customer for food

Menu| List of available food items

Inventory| Collection of available ingredients/items

Bill| Amount payable by the customer

Database| Organized collection of system data

1.4 References

- IEEE Recommended Practice for Software Requirements Specifications

- Database Management System concepts
- Software Engineering principles

1.5 Overview

This document contains the overall description, functional requirements, non-functional requirements, external interfaces, system constraints, assumptions, and dependencies of the Restaurant Management System

2. Overall Description

2.1 Product Perspective

The Restaurant Management System is a standalone computerized application that manages restaurant operations.

The system connects different activities such as:

Customer → Order → Kitchen → Billing → Payment → Reports

It stores information in a database and allows authorized users to access and update the required information.

2.2 Product Functions

The major functions of the system are:

1. User Login
2. Customer Management

3. Table Management
4. Menu Management
5. Food Ordering
6. Kitchen Order Management
7. Billing
8. Payment Management
9. Inventory Management
10. Employee Management
11. Sales Management
12. Report Generation

2.3 User Characteristics

Admin / Manager

- Manages the complete restaurant system.
- Adds, updates, and deletes menu items.
- Manages employees and inventory.
- Views sales and other reports.

Cashier

- Creates customer orders.
- Generates bills.
- Records payments.
- Updates order status.

Kitchen Staff

- Views incoming food orders.
- Updates food preparation status.
- Marks orders as ready.

Customer

- Views the menu.
- Selects food items.
- Places orders.
- Receives the bill.

2.4 Operating Environment

The system can operate in:

Hardware

- Computer/Laptop
- Keyboard and mouse
- Printer
- Local network
- Server/database system

Software

- Windows/Linux operating system
- Web browser or application interface
- Database Management System
- Application server

2.5 Constraints

- Only authorized users can access administrative functions.
- The system requires a working database.
- User permissions must be maintained.
- Accurate data entry is required.
- The system should maintain customer and transaction data securely.

2.6 Assumptions and Dependencies

- Users have basic computer knowledge.
- The database server is available.
- Restaurant employees are trained to use the system.
- Menu and inventory information is updated regularly.
- Required hardware such as computers and printers is available

3. Specific Requirements

3.1 Functional Requirements

FR1 – User Login

The system shall provide a login facility.

- User enters username and password.
- System validates the credentials.
- Authorized users can access their respective functions.
- Unauthorized users shall be denied access.

FR2 – Customer Management

The system shall allow the user to:

- Add customer details.
- View customer details.
- Update customer information.
- Delete customer information when required.

FR3 – Table Management

The system shall manage restaurant tables.

- Display available tables.
- Display occupied tables.
- Assign tables to customers.
- Update table status after an order is completed.

FR4 – Menu Management

The admin shall be able to:

- Add food items.
- Update food item details.
- Delete food items.
- Set food prices.
- Change item availability.
- Display food categories.

FR5 – Food Ordering

The system shall allow users to create food orders.

- Select food items.
- Specify quantity.
- Calculate the total order amount.
- Generate an order number.
- Store order information in the database.

FR6 – Kitchen Management

The system shall send orders to the kitchen.

Kitchen staff can:

- View new orders.
- View ordered food items.
- Update preparation status.
- Mark orders as ready.

FR7 – Billing

The system shall automatically generate bills.

The bill shall contain:

- Order number
- Food items
- Quantity
- Price
- Total amount
- Applicable taxes
- Final payable amount

FR8 – Payment Management

The system shall record customer payments.

Payment information may include:

- Cash payment
- Card payment
- Digital payment

After successful payment, the order shall be marked as completed.

FR9 – Inventory Management

The system shall manage restaurant stock.

- Add inventory items.
- Update stock quantity.
- Delete inventory items.
- Monitor available stock.
- Display low-stock items.

FR10 – Employee Management

The admin shall be able to:

- Add employee details.
- Update employee information.
- Delete employee records.
- View employee details.
- Assign employee roles.

FR11 – Report Generation

The system shall generate reports such as:

- Daily sales report
- Monthly sales report
- Food sales report
- Inventory report
- Employee report
- Order report

FR12 – Data Management

The system shall store all important information in a database and allow authorized users to retrieve and update the data

4. External Interface Requirements

4.1 User Interface

The system shall provide a simple and user-friendly interface.

The interface shall contain:

- Login page
- Dashboard
- Menu page
- Customer page

- Order page
- Billing page
- Inventory page
- Employee page
- Reports page

4.2 Hardware Interface

The system may interact with:

- Computer/Laptop
- Printer
- Barcode scanner, if required
- Network devices

4.3 Software Interface

The system may interact with:

- Operating system
- Database Management System
- Web browser
- Payment processing software, if applicable

4.4 Communication Interface

The system shall use a local network or internet connection where required for communication between application components and the database

5. Non-Functional Requirements

5.1 Performance

- The system should respond quickly to user requests.
- Orders should be processed without unnecessary delay.
- Reports should be generated efficiently.
- The database should support multiple transactions.

5.2 Security

- User authentication shall be provided.
- Passwords shall be protected.
- Users shall have role-based access.
- Only authorized users can access sensitive information.

5.3 Reliability

- The system should provide accurate results.
- Data should not be lost during normal operation.
- Regular database backup should be supported.

5.4 Usability

- The system should be easy to learn.
- The interface should be simple and clear.
- Error messages should be understandable.
- Users should be able to perform common tasks easily.

5.5 Maintainability

- The software should be easy to maintain.
- Menu items and prices should be easily updated.
- Future features should be added without major changes.

5.6 Scalability

The system should support an increase in

- Number of customers
- Number of orders
- Number of employees
- Number of menu items
- Amount of stored data

5.7 Availability

The system should be available during restaurant operating hours with minimum downtime

6. Database Requirements

The system shall maintain a database containing the following major tables:

1. User
2. Customer
3. Employee
4. Table
5. Menu
6. Order
7. Order Details
8. Payment
9. Inventory
10. Reports

Important Data

Customer: Customer ID, Name, Phone Number

Employee: Employee ID, Name, Role, Phone Number

Menu: Food ID, Food Name, Category, Price, Availability

Order: Order ID, Customer ID, Table ID, Order Date, Status, Total Amount

Payment: Payment ID, Order ID, Payment Mode, Amount, Payment Date

Inventory: Item ID, Item Name, Quantity, Unit, Reorder Level

7. System Workflow

The basic workflow of the system is

Login

↓

Select Table / Customer



View Menu



Place Order



Send Order to Kitchen



Prepare Food



Update Order Status



Generate Bill



Make Payment



Complete Order



Update Inventory



Generate Reports

8. Benefits of the System

- Reduces manual work.
- Saves time.
- Reduces billing errors.
- Improves order management.

- Provides accurate sales information.
- Helps manage inventory.
- Improves customer service.
- Provides quick access to reports.
- Maintains centralized data

9. Conclusion

The Restaurant Management System provides an efficient way to manage restaurant operations. It integrates customer management, food ordering, kitchen management, billing, payment, inventory, employee management, and reporting into a single system.

The system reduces manual effort, improves accuracy, and provides quick access to important restaurant information.